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2WD Drive Autonomous Mobile Robot Vehicle

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Introduction

Autonomous mobile robots are a key focus in the study of embedded systems and robotics because they bring together mechanical design, electronics, and control theory into a single, working device. In this project, we designed and built such a robot to take part in the Autonomous Mobile Robots Challenge, an event closely aligned with the goals of the Embedded Systems curriculum. The competition tests the robot's ability to navigate independently through a multi-zone course, requiring it to complete each stage accurately and efficiently.

The course itself is divided into several distinct sections, including a line-following track, a dark tunnel, a maze, and a final parking area. Each zone presents its own unique challenges, simulating real-world conditions that push the robot's navigation and decision-making systems to perform reliably under varying and unpredictable circumstances.

Design

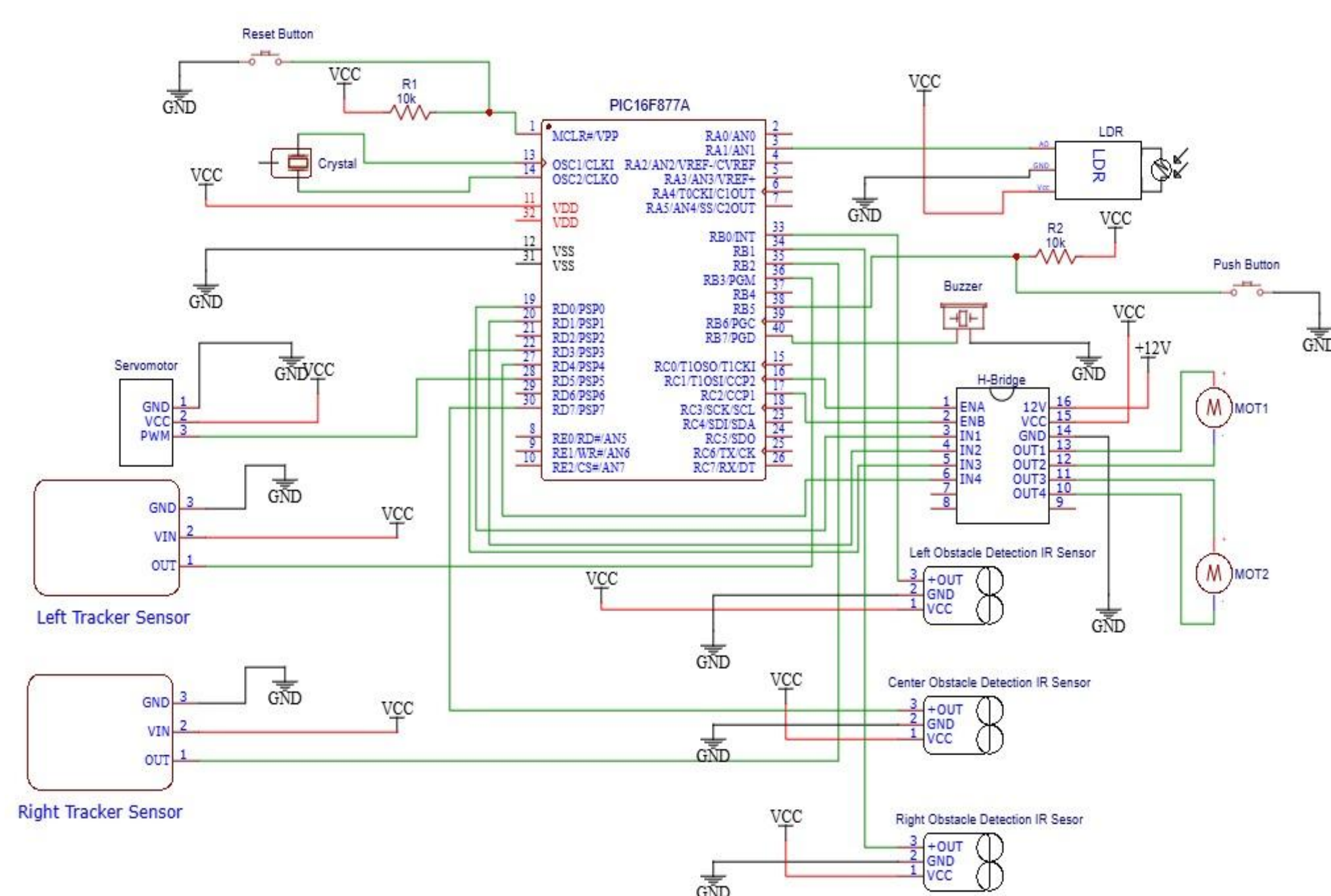


Figure 1: Electrical Design

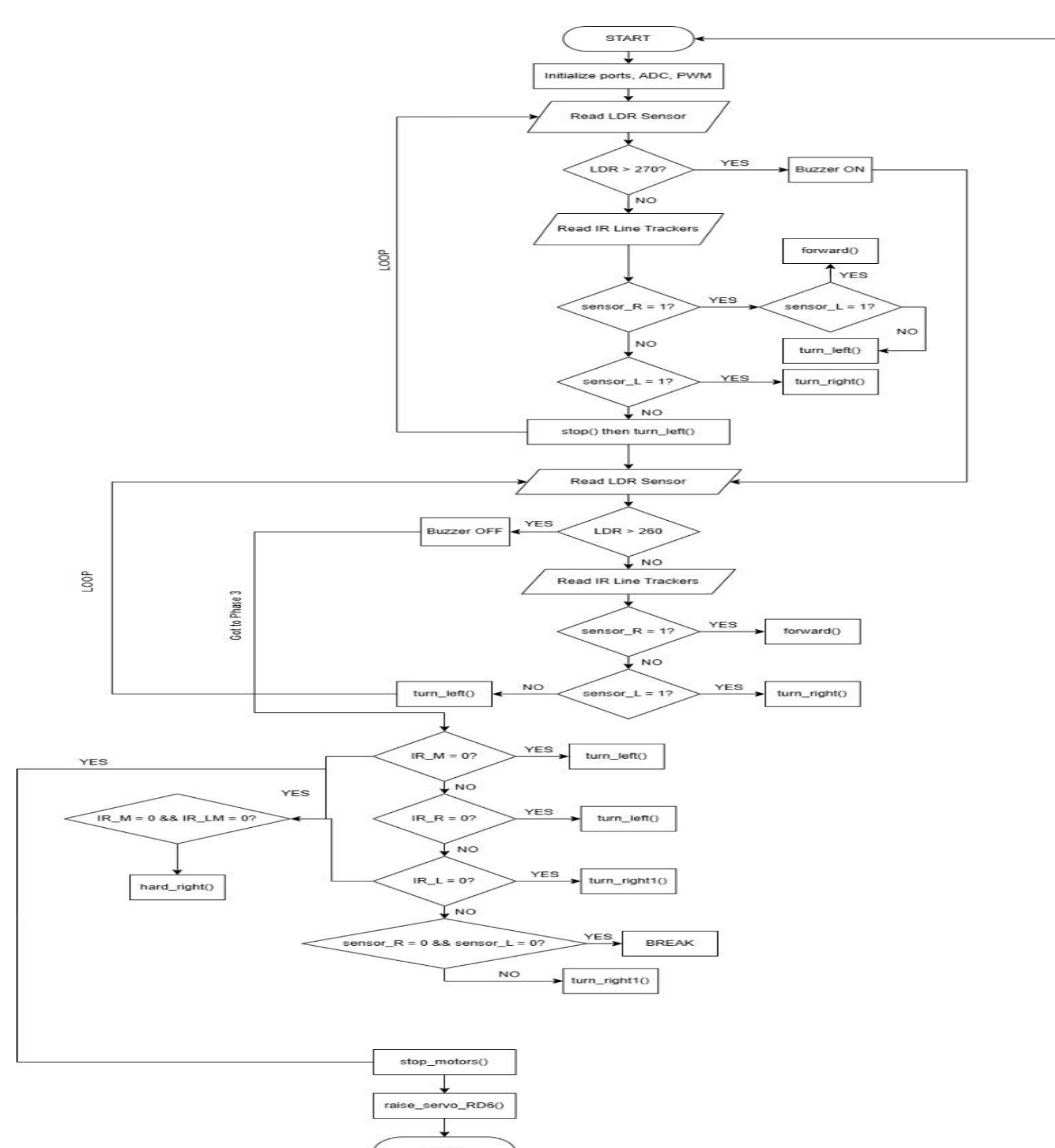


Figure 2: Software Design

Results

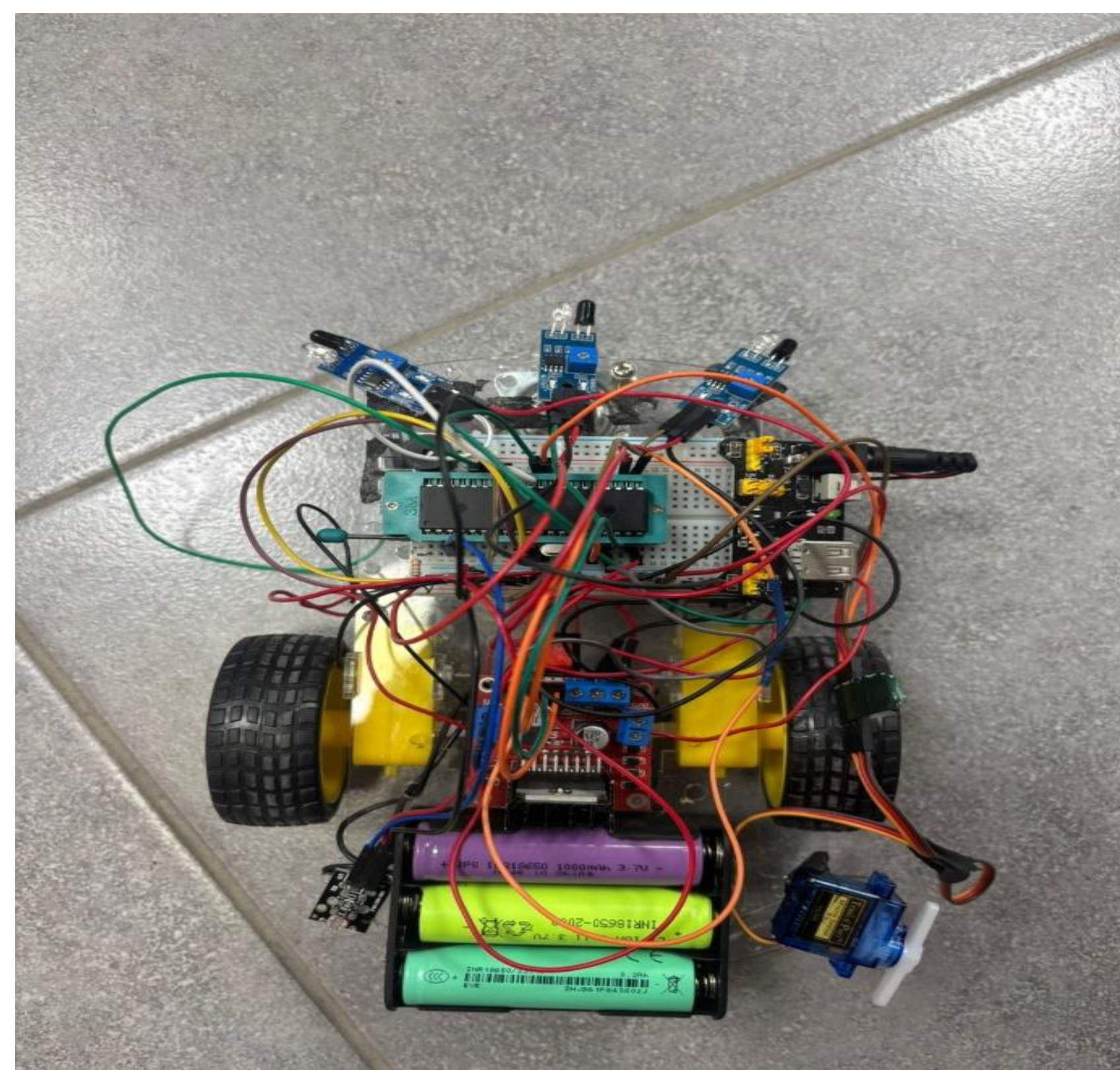
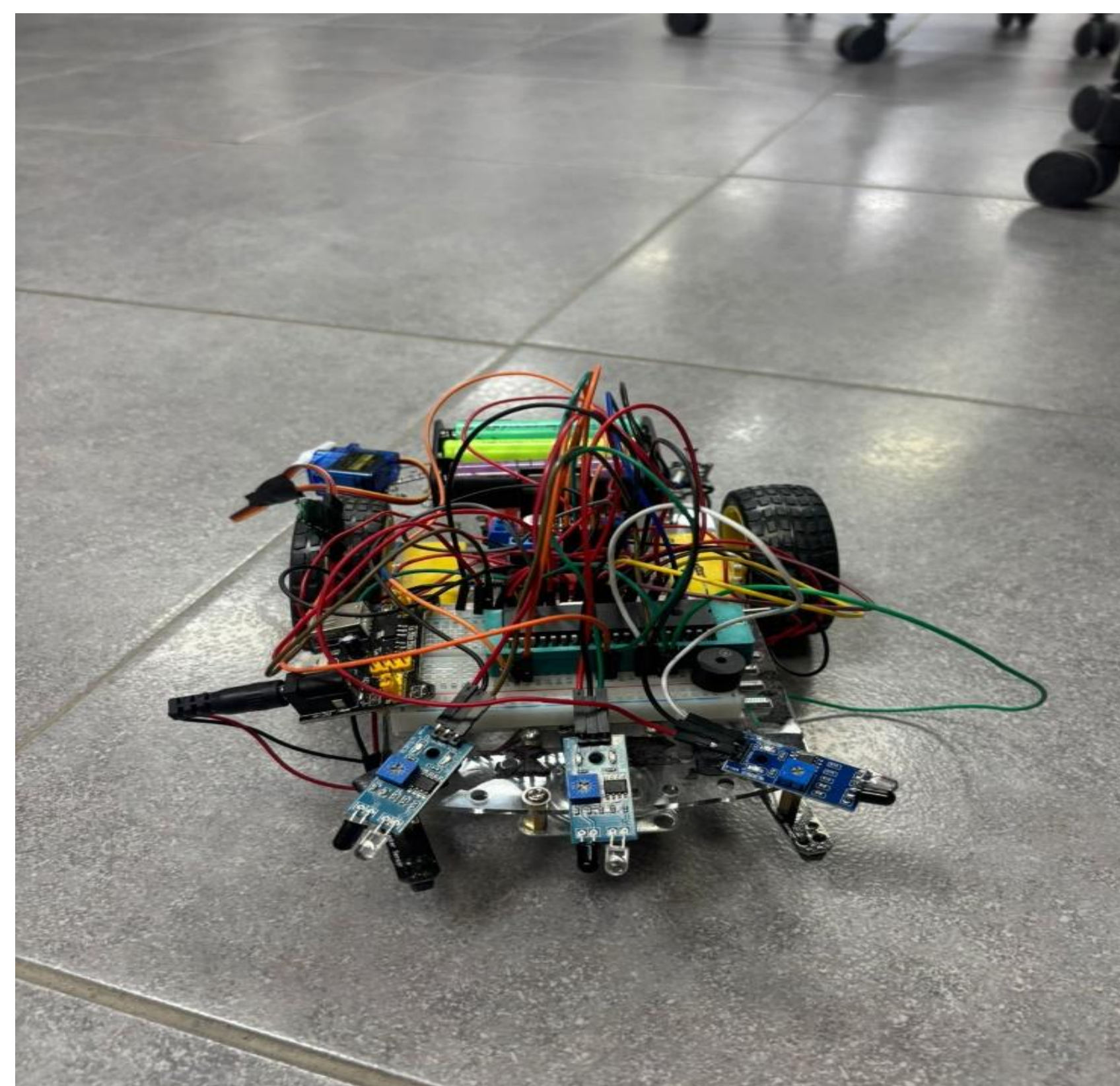


Figure 4: Mechanical Design

The images shown above display that the robot is built on a two-wheel differential drive (2WD) chassis with two DC motors directly attached to the wheels, providing efficient propulsion and precise control. This design allows the robot to turn and maneuver accurately by adjusting the speed of each motor independently.

Conclusion

In conclusion, the development of the vehicle integrates mechanical, electrical and embedded programming into a single autonomous system. Through careful implementation of the entire track, the robot successfully navigates the designated track and responds to environmental cues. The robot's behavior is governed by a state-based embedded software architecture that processes sensor inputs and timing events to control navigation decisions across all track zones. This project provides relevant experience in robotics and problem-solving.